

Software & Observatory — Study Notes

CASSA RA 2026-27 · companion to the practice sheet & surveys notes

*Your SOP says you can contribute across the **astrophysics, observatory, and software** sides of CASSA. You are applying to Astrophysics, but the panel may probe the other two. This sheet gives you enough to answer confidently and tie each back to work you have actually done.*

PART A — RA SOFTWARE

A1. The Python astronomy stack (name these fluently)

- **NumPy / SciPy** — arrays, linear algebra, optimization, integration (root-finding, ODEs).
- **Astropy** — the core library: **units** & **constants**, **coordinates** (RA/Dec, frame transforms), **FITS I/O**, **WCS** (pixel ↔ sky), time, cosmology calculators.
- **Matplotlib** — plotting; **Pandas** — tabular data.
- **Astroquery** — programmatic access to archives (VizieR, MAST, SDSS, NED, Simbad).

A2. SED fitting & Bayesian sampling (your daily tools)

- **python-fsps** — Flexible Stellar Population Synthesis (the model engine behind Rose et al. 2019 and your pipeline).
- Samplers: **emcee** (MCMC ensemble), **dynesty** (dynamic nested), **PyMultiNest** (MultiNest wrapper, uses **MPI**), **nautilus-sampler** (NN-boosted importance nested sampling).
- Higher-level SED codes: **Prospector**, **BAGPIPES** (you know bagpipes from your own work).
- **corner** — posterior corner plots.

A3. Data formats & archives

- **FITS** (Flexible Image Transport System) — the standard astro format: ASCII **header** (keywords) + binary **data** (image or table). Know: HDU, header cards, WCS keywords.
- **WCS** — World Coordinate System, maps detector pixels to sky coordinates.
- Archives you should be able to name: **MAST** (HST/JWST/TESS), **SDSS SkyServer / CasJobs**, **IRSA**, **VizieR / CDS**, **NED**, **ADS** (literature).

A4. Photometry / image tools

- **photutils** (Astropy) — aperture & PSF photometry, source detection.
- **Source Extractor (SExtractor)**, **DAOPHOT** — classic source detection/photometry.
- **SWarp / Montage** — image coaddition/mosaicking; **astrometry.net** — blind plate solving.

A5. Machine learning (your strength)

- **scikit-learn** (pipelines, metrics), **LightGBM / XGBoost** (gradient-boosted trees — your **PIE-ADA** used LightGBM), **TensorFlow / PyTorch** (deep nets).
- Architectures you have used: **CNN** (RGC radio-morphology classifier), **CVAE + Transformer** (LunaXNet lunar mineralogy).
- Metrics: **weighted log-loss** (PIE-ADA), accuracy, precision/recall, ROC-AUC, confusion matrix.
- **Transient brokers** that will process LSST alerts: **ANTARES**, **ALeRCE**, **Lasair**, **Fink** — this is where PIE-ADA-style classifiers get deployed.

A6. HPC, reproducibility, deployment

- **SLURM** — cluster job scheduler (submit, array jobs, resource limits). Your SN runs used **automated SLURM** on the **Timaeus** HPC cluster. **MPI** for MultiNest parallelism.
- **git / GitHub** — version control; open pipeline **snhostage** is your reproducibility contribution (Paper 3).
- **conda / virtualenv** — reproducible environments; containers (Docker/Singularity) on clusters.
- **Web / apps: Streamlit** (your **CANDLE** interactive SED-fitting web tool), **Firebase + GitHub Pages (knackbook)**, **Kaggle** kernels (LunaXNet training).

A7. Software viva bridges (say these)

- "What have you built?" → **snhostage** pipeline + **CANDLE** Streamlit app (SN host SED fitting), **PIE-ADA** classifier, **RGC** data-curation code, **knackbook** web app.
- "How do you handle big runs?" → **SLURM** job arrays on **Timaeus**, **Nautilus** to cut likelihood calls ~60x, **MPI** for MultiNest.
- "How is your work reproducible?" → open-source pipeline, fixed priors/seeds, versioned on **GitHub**, shared **README/Overleaf**.

PART B — OBSERVATORY

B1. Coordinates & timekeeping

- **Equatorial system (RA, Dec)** — fixed to the sky; RA in hours (0-24h), Dec in degrees ($\pm 90^\circ$).
- **Horizontal system (Altitude, Azimuth)** — local to observer & time; Alt = angle above horizon, Az = compass bearing.
- **Sidereal time** — star time; an object crosses the meridian when Local Sidereal Time = its RA. **Hour angle** = LST – RA.
- **Airmass** = $\sec(z) \approx 1/\cos(z)$, z = zenith angle. Airmass = 1 at zenith; larger near horizon → more **atmospheric extinction** (objects dimmer, redder low in the sky).

B2. Telescope mounts (you have hands-on experience here)

- **Alt-azimuth mount** — two axes: **altitude** (up/down) + **azimuth** (left/right). Simpler, cheaper, compact. Downsides: needs **two motors** to track + software, and suffers **field rotation** (long exposures need a de-rotator). **Dobsonian** = simple alt-az. *You have strong practical experience on alt-az mounts.*
- **Equatorial mount** — one axis (**polar/RA axis**) aligned to the **celestial pole**; then a **single motor** tracks the sky, no field rotation. **GEM** = German Equatorial Mount. Needs **polar alignment**. *You have working knowledge of these.*
- **UniStellar (eVscope)** — automated **smart alt-az** telescope: **plate-solving auto-alignment**, electronic eyepiece, stacks frames live. Used for **citizen-science campaigns** (asteroid occultations, exoplanet transits, comets, planetary defense). *You cite it in your SOP as an automated system you know.*

B3. Telescope optics

- **Refractor** — lens objective (chromatic aberration issue). **Reflector** — mirror: **Newtonian**, **Cassegrain**, **Ritchey-Chrétien** (pro standard, e.g. HST), **Schmidt-Cassegrain / Maksutov** (compact catadioptric).
- Key numbers: **Aperture D** (light gathering $\propto D^2$, bigger = fainter objects), **focal length f**, **focal ratio f/# = f/D** (smaller = "faster", wider field), **plate scale** (arcsec/mm or arcsec/pixel).

- **Angular resolution (diffraction limit):** $\theta \approx 1.22 \lambda / D$ (radians) — bigger aperture → finer detail. In practice ground-based resolution is limited by **seeing** (~1 arcsec, atmospheric turbulence), not diffraction.
- **Adaptive optics (AO)** — deformable mirror corrects seeing in real time (with a guide star). **Active optics** — slow correction of mirror shape/gravity sag.

B4. Detectors

- **CCD vs CMOS** — CMOS now common (faster readout, lower power); CCD long the scientific standard.
- Detector properties: **Quantum Efficiency (QE)**, **read noise**, **dark current**, **gain** (e⁻/ADU), **full-well capacity**, **linearity**, **saturation**.
- Signal-to-noise: **SNR** = $S / \sqrt{S + n_{\text{pix}}(B + D + R^2)}$ (source + sky + dark + read noise) — the basis of exposure-time calculators.

B5. Calibration (essential — likely asked)

- **Bias frame** — zero-second exposure, removes electronic offset.
- **Dark frame** — same exposure time, shutter closed, removes thermal/dark current.
- **Flat field** — uniform light (twilight/dome), corrects pixel-to-pixel sensitivity + vignetting + dust.
- Reduced image = **(raw – bias – dark) / flat**.

B6. Photometry & filters

- **Aperture photometry** (sum flux in a circle, subtract sky annulus) vs **PSF photometry** (fit the point-spread function — for crowded fields).
- **Zero point & standard stars** calibrate instrumental → real magnitudes; **color terms** correct filter mismatch; **extinction coefficient** corrects for airmass.
- **Filter systems:** **Johnson-Cousins UBVRI**, **SDSS ugriz**, **2MASS JHKs** (near-IR). (Ties to your ugriz / ugrizYJH SED work.)
- **Magnitude** is logarithmic: $m_1 - m_2 = -2.5 \log_{10}(F_1/F_2)$; brighter = smaller number.

B7. Observing operations

- **Proposals & scheduling:** classical (assigned nights) vs **queue/service** (staff observe for you) vs **target-of-opportunity (ToO)** — ToO is how **transients/supernovae** get rapid follow-up (connects to your SN work).
- **Observing planning:** target visibility windows, airmass charts, moon phase/separation, seeing, twilight.
- **Site quality:** dark sky, high & dry (low water vapour for IR), stable atmosphere — why Chile (Las Campanas/Cerro Pachón) hosts CSP, DES, Rubin.

B8. Observatory viva bridges (say these)

- "What is your hands-on experience?" → operated telescopes on **alt-az mounts** (strong practical), working knowledge of **equatorial** mounts and **automated UniStellar** systems.
- "Why does airmass/seeing matter?" → they set how faint and how sharp you can go; drive when/where to observe.
- "How is a raw image turned into science?" → bias/dark/flat calibration → source detection → aperture/PSF photometry → zero-point calibration → magnitudes/SED (my SED fitting starts from that catalogue).

Quick-recall (flash-card)

Software: Astropy (units/coords/FITS/WCS) · python-fsps · emcee/dynesty/PyMultiNest(MPI)/nautilus · Prospector/BAGPIPES · photutils/SExtractor · LightGBM (PIE-ADA) · SLURM on Timaeus · git · Streamlit

(CANDLE) · Firebase+GitHub Pages (knackbook) · brokers ANTARES/ALeRCE/Lasair. **Observatory:** RA/Dec vs Alt/Az · sidereal time, hour angle · airmass = $\sec(z) \rightarrow$ extinction · alt-az (field rotation, 2 motors) vs equatorial (1 motor, polar align) · UniStellar smart alt-az · resolution $\theta \approx 1.22\lambda/D$ vs seeing $\sim 1''$ · AO · CCD/CMOS (QE, read noise, gain) · bias/dark/flat: $(\text{raw} - \text{bias} - \text{dark})/\text{flat}$ · aperture vs PSF photometry · zero point + standard stars · UBVRI/ugriz/JHK · ToO for transients.

Likely viva Qs

1. Difference between alt-az and equatorial mounts? Which have you used?
2. What is airmass and why does it matter?
3. What calibration frames do you take and what does each remove?
4. What software do you use for SED fitting and for running jobs at scale?
5. Name astronomy data archives / a transient broker.
6. What is the diffraction limit, and why don't ground telescopes reach it?